

Experiment 3

Aim :

To create an Employee–Department–Project database schema, insert at least 30 employees across 5 departments and 8 projects, and execute SQL queries demonstrating selection, projection, aggregation, GROUP BY, HAVING, CASE expressions, and ORDER BY.

Description :

This experiment demonstrates the use of SQL for managing and analyzing data stored in an Employee–Department–Project schema. The database contains five departments, eight projects, and at least thirty employees distributed among the departments and projects.

The experiment covers important SQL operations including selection using WHERE, projection using SELECT, aggregation using functions such as COUNT, AVG, SUM, and MAX, grouping records using GROUP BY, filtering grouped results using HAVING, conditional classification using CASE expressions, and sorting query results using ORDER BY.

The schema uses primary keys and foreign keys to maintain relationships between departments, employees, and projects. The queries help generate useful reports such as department-wise employee counts, average salaries, project assignments, salary classifications, and departments satisfying specified conditions.

Source Code :

```
-- Create Department table
CREATE TABLE Department (
    dept_id INT PRIMARY KEY,
    dept_name VARCHAR(50) NOT NULL,
    location VARCHAR(50)
);

-- Create Project table
CREATE TABLE Project (
    project_id INT PRIMARY KEY,
    project_name VARCHAR(100) NOT NULL,
    budget DECIMAL(12,2),
    dept_id INT,
    FOREIGN KEY (dept_id) REFERENCES Department(dept_id)
);

-- Create Employee table
CREATE TABLE Employee (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(100) NOT NULL,
    email VARCHAR(100),
    salary DECIMAL(10,2),
    hire_date DATE,
    dept_id INT,
    project_id INT,
    FOREIGN KEY (dept_id) REFERENCES Department(dept_id),
    FOREIGN KEY (project_id) REFERENCES Project(project_id)
);
```

```

-- Insert 5 departments
INSERT INTO Department VALUES
(1, 'Computer Science', 'Block A'),
(2, 'Human Resources', 'Block B'),
(3, 'Finance', 'Block C'),
(4, 'Marketing', 'Block D'),
(5, 'Operations', 'Block E');

-- Insert 8 projects
INSERT INTO Project VALUES
(101, 'AI Research', 500000.00, 1),
(102, 'Web Development', 350000.00, 1),
(103, 'Recruitment Portal', 180000.00, 2),
(104, 'Financial Analytics', 450000.00, 3),
(105, 'Digital Marketing', 300000.00, 4),
(106, 'Inventory System', 250000.00, 5),
(107, 'Cloud Migration', 600000.00, 1),
(108, 'Business Automation', 400000.00, 5);

-- Insert 30 employees
INSERT INTO Employee VALUES
(1001, 'Aarav Sharma', 'aarav@company.com', 65000, '2022-01-15', 1, 101),
(1002, 'Riya Verma', 'riya@company.com', 72000, '2021-03-20', 1, 102),
(1003, 'Arjun Singh', 'arjun@company.com', 58000, '2023-02-10', 1, 101),
(1004, 'Neha Gupta', 'neha@company.com', 81000, '2020-06-18', 1, 107),
(1005, 'Kabir Mehta', 'kabir@company.com', 69000, '2022-08-12', 1, 102),
(1006, 'Ananya Rao', 'ananya@company.com', 76000, '2021-11-05', 1, 107),

(1007, 'Rahul Kumar', 'rahul@company.com', 48000, '2022-04-11', 2, 103),
(1008, 'Simran Kaur', 'simran@company.com', 52000, '2023-01-25', 2, 103),
(1009, 'Vikram Joshi', 'vikram@company.com', 55000, '2021-09-17', 2, 103),
(1010, 'Pooja Nair', 'pooja@company.com', 61000, '2020-12-01', 2, 103),
(1011, 'Karan Malhotra', 'karan@company.com', 50000, '2022-07-22', 2, 103),
(1012, 'Ishita Jain', 'ishita@company.com', 57000, '2023-05-15', 2, 103),

(1013, 'Aditya Patel', 'aditya@company.com', 68000, '2021-02-14', 3, 104),
(1014, 'Meera Shah', 'meera@company.com', 75000, '2020-10-19', 3, 104),
(1015, 'Rohan Das', 'rohan@company.com', 63000, '2022-03-08', 3, 104),
(1016, 'Sneha Iyer', 'sneha@company.com', 71000, '2021-07-13', 3, 104),
(1017, 'Manish Yadav', 'manish@company.com', 59000, '2023-04-20', 3, 104),
(1018, 'Tanya Kapoor', 'tanya@company.com', 82000, '2019-11-30', 3, 104),

(1019, 'Nikhil Bansal', 'nikhil@company.com', 54000, '2022-02-21', 4, 105),
(1020, 'Kritika Sethi', 'kritika@company.com', 62000, '2021-05-16', 4,
105),
(1021, 'Yash Thakur', 'yash@company.com', 57000, '2023-03-11', 4, 105),
(1022, 'Aditi Roy', 'aditi@company.com', 66000, '2020-08-09', 4, 105),
(1023, 'Mohit Arora', 'mohit@company.com', 59000, '2022-10-14', 4, 105),
(1024, 'Diya Khanna', 'diya@company.com', 73000, '2021-12-18', 4, 105),

(1025, 'Sahil Verma', 'sahil@company.com', 56000, '2022-01-07', 5, 106),
(1026, 'Muskan Singh', 'muskan@company.com', 64000, '2020-04-23', 5, 108),
(1027, 'Harsh Vardhan', 'harsh@company.com', 61000, '2021-06-30', 5, 106),
(1028, 'Nisha Sharma', 'nisha@company.com', 70000, '2019-09-12', 5, 108),
(1029, 'Dev Agarwal', 'dev@company.com', 53000, '2023-02-28', 5, 106),
(1030, 'Ira Saxena', 'ira@company.com', 78000, '2020-11-16', 5, 108);

-- 1. Selection: display employees with salary greater than 70000
SELECT *

```

```
FROM Employee
WHERE salary > 70000;
```

The screenshot shows the ByteXL SQL IDE interface. The left pane displays a SQL query file named 'queries.sql' with the following content:

```

78 (1010, 'Pooja Nair', 'pooja@company.com', 61000, '2020-12-01', 2, 103),
79 (1011, 'Karan Malhotra', 'karan@company.com', 50000, '2022-07-22', 2, 103),
80 (1012, 'Ishita Jain', 'ishita@company.com', 57000, '2023-05-15', 2, 103),
81
82 (1013, 'Aditya Patel', 'aditya@company.com', 68000, '2021-02-14', 3, 104),
83 (1014, 'Meera Shah', 'meera@company.com', 75000, '2020-10-19', 3, 104),
84 (1015, 'Rohan Das', 'rohan@company.com', 63000, '2022-03-08', 3, 104),
85 (1016, 'Sneha Iyer', 'sneha@company.com', 71000, '2021-07-13', 3, 104),
86 (1017, 'Manish Yadav', 'manish@company.com', 59000, '2023-04-20', 3, 104),
87 (1018, 'Tanya Kapoor', 'tanya@company.com', 82000, '2019-11-30', 3, 104),
88
89 (1019, 'Nikhil Bansal', 'nikhil@company.com', 54000, '2022-02-21', 4, 105),
90 (1020, 'Kritika Sethi', 'kritika@company.com', 62000, '2021-05-16', 4, 105),
91 (1021, 'Yash Thakur', 'yash@company.com', 57000, '2023-03-11', 4, 105),
92 (1022, 'Aditi Roy', 'aditi@company.com', 66000, '2020-08-09', 4, 105),
93 (1023, 'Mohit Arora', 'mohit@company.com', 59000, '2022-10-14', 4, 105),
94 (1024, 'Diya Khanna', 'diya@company.com', 73000, '2021-12-18', 4, 105),
95
96 (1025, 'Sahil Verma', 'sahil@company.com', 56000, '2022-01-07', 5, 106),
97 (1026, 'Muskan Singh', 'muskan@company.com', 64000, '2020-04-23', 5, 106),
98 (1027, 'Harsh Vardhan', 'harsh@company.com', 61000, '2021-06-30', 5, 106),
99 (1028, 'Nisha Sharma', 'nisha@company.com', 70000, '2019-09-12', 5, 106),
100 (1029, 'Dev Agarwal', 'dev@company.com', 53000, '2023-02-28', 5, 106),
101 (1030, 'Ira Saxena', 'ira@company.com', 78000, '2020-11-16', 5, 106);
102
103
104 -- *****
105 -- QUERY 1: SELECTION
106 -- Display employees with salary greater than 70000
107 -- *****
108
109 SELECT *
110 FROM Employee
111 WHERE salary > 70000;
112
113

```

The right pane shows the execution output for the query. The output is a table with the following columns: emp_id, emp_name, email, salary, hire_date, dept_id, project_id. The output shows 10 rows of data, all with salaries greater than 70000.

emp_id	emp_name	email	salary	hire_date	dept_id	project_id
1002	Riya Verma	riya@company.com	72000.00	2021-03-20	1	102
1004	Neha Gupta	neha@company.com	81000.00	2020-06-18	1	107
1006	Ananya Rao	ananya@company.com	76000.00	2021-11-05	1	107
1014	Meera Shah	meera@company.com	75000.00	2020-10-19	3	104
1016	Sneha Iyer	sneha@company.com	71000.00	2021-07-13	3	104
1018	Tanya Kapoor	tanya@company.com	82000.00	2019-11-30	3	104
1024	Diya Khanna	diya@company.com	73000.00	2021-12-18	4	105
1030	Ira Saxena	ira@company.com	78000.00	2020-11-16	5	106

-- 2. Projection: display employee names and salaries

```
SELECT emp_name, salary
FROM Employee;
```

The screenshot shows the ByteXL SQL IDE interface. The left pane displays a SQL query file named 'queries.sql' with the following content:

```

71 (1004, 'Neha Gupta', 'neha@company.com', 81000, '2020-06-18', 1, 107),
72 (1005, 'Kabir Mehta', 'kabir@company.com', 69000, '2022-08-12', 1, 102),
73 (1006, 'Ananya Rao', 'ananya@company.com', 76000, '2021-11-05', 1, 107),
74
75 (1007, 'Rahul Kumar', 'rahul@company.com', 48000, '2022-04-11', 2, 103),
76 (1008, 'Simran Kaur', 'simran@company.com', 52000, '2023-01-25', 2, 103),
77 (1009, 'Vikram Joshi', 'vikram@company.com', 55000, '2021-09-17', 2, 103),
78 (1010, 'Pooja Nair', 'pooja@company.com', 61000, '2020-12-01', 2, 103),
79 (1011, 'Karan Malhotra', 'karan@company.com', 50000, '2022-07-22', 2, 103),
80 (1012, 'Ishita Jain', 'ishita@company.com', 57000, '2023-05-15', 2, 103),
81
82 (1013, 'Aditya Patel', 'aditya@company.com', 68000, '2021-02-14', 3, 104),
83 (1014, 'Meera Shah', 'meera@company.com', 75000, '2020-10-19', 3, 104),
84 (1015, 'Rohan Das', 'rohan@company.com', 63000, '2022-03-08', 3, 104),
85 (1016, 'Sneha Iyer', 'sneha@company.com', 71000, '2021-07-13', 3, 104),
86 (1017, 'Manish Yadav', 'manish@company.com', 59000, '2023-04-20', 3, 104),
87 (1018, 'Tanya Kapoor', 'tanya@company.com', 82000, '2019-11-30', 3, 104),
88
89 (1019, 'Nikhil Bansal', 'nikhil@company.com', 54000, '2022-02-21', 4, 105),
90 (1020, 'Kritika Sethi', 'kritika@company.com', 62000, '2021-05-16', 4, 105),
91 (1021, 'Yash Thakur', 'yash@company.com', 57000, '2023-03-11', 4, 105),
92 (1022, 'Aditi Roy', 'aditi@company.com', 66000, '2020-08-09', 4, 105),
93 (1023, 'Mohit Arora', 'mohit@company.com', 59000, '2022-10-14', 4, 105),
94 (1024, 'Diya Khanna', 'diya@company.com', 73000, '2021-12-18', 4, 105),
95
96 (1025, 'Sahil Verma', 'sahil@company.com', 56000, '2022-01-07', 5, 106),
97 (1026, 'Muskan Singh', 'muskan@company.com', 64000, '2020-04-23', 5, 106),
98 (1027, 'Harsh Vardhan', 'harsh@company.com', 61000, '2021-06-30', 5, 106),
99 (1028, 'Nisha Sharma', 'nisha@company.com', 70000, '2019-09-12', 5, 106),
100 (1029, 'Dev Agarwal', 'dev@company.com', 53000, '2023-02-28', 5, 106),
101 (1030, 'Ira Saxena', 'ira@company.com', 78000, '2020-11-16', 5, 106);
102
103
104 -- *****
105 -- QUERY 2: PROJECTION
106 -- Display employee names and salaries
107 -- *****
108
109 SELECT emp_name, salary
110 FROM Employee;
111
112

```

The right pane shows the execution output for the query. The output is a table with the following columns: emp_name, salary. The output shows 30 rows of data, all with salaries greater than 70000.

emp_name	salary
Aarav Sharma	65000.00
Riya Verma	72000.00
Arjun Singh	58000.00
Neha Gupta	81000.00
Kabir Mehta	69000.00
Ananya Rao	76000.00
Rahul Kumar	48000.00
Simran Kaur	52000.00
Vikram Joshi	55000.00
Pooja Nair	61000.00
Karan Malhotra	50000.00
Ishita Jain	57000.00
Aditya Patel	68000.00
Meera Shah	75000.00
Rohan Das	63000.00
Sneha Iyer	71000.00
Manish Yadav	59000.00
Tanya Kapoor	82000.00
Nikhil Bansal	54000.00
Kritika Sethi	62000.00
Yash Thakur	57000.00
Aditi Roy	66000.00
Mohit Arora	59000.00
Diya Khanna	73000.00
Sahil Verma	56000.00
Muskan Singh	64000.00
Harsh Vardhan	61000.00
Nisha Sharma	70000.00
Dev Agarwal	53000.00
Ira Saxena	78000.00

-- 3. Aggregation: count employees and calculate average salary

```
SELECT COUNT(*) AS total_employees,
       AVG(salary) AS average_salary
FROM Employee;
```

byteXL Ctrl + K NISHANT KUMAR 88 ms

queries.sql x MySQL Run

```

79 (1011, 'Karan Malhotra', 'karan@company.com', 50000, '2022-07-22', 2, 103),
80 (1012, 'Ishita Jain', 'ishita@company.com', 57000, '2023-05-15', 2, 103),
81
82 (1013, 'Aditya Patel', 'aditya@company.com', 68000, '2021-02-14', 3, 104),
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84 (1015, 'Rohan Das', 'rohan@company.com', 63000, '2022-03-08', 3, 104),
85 (1016, 'Sneha Iyer', 'sneha@company.com', 71000, '2021-07-13', 3, 104),
86 (1017, 'Manish Yadav', 'manish@company.com', 59000, '2023-04-20', 3, 104),
87 (1018, 'Tanya Kapoor', 'tanya@company.com', 82000, '2019-11-30', 3, 104),
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89 (1019, 'Nikhil Bansal', 'nikhil@company.com', 54000, '2022-02-21', 4, 105),
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91 (1021, 'Yash Thakur', 'yash@company.com', 57000, '2023-03-11', 4, 105),
92 (1022, 'Aditi Roy', 'aditi@company.com', 66000, '2020-08-09', 4, 105),
93 (1023, 'Mohit Arora', 'mohit@company.com', 59000, '2022-10-14', 4, 105),
94 (1024, 'Diya Khanna', 'diya@company.com', 73000, '2021-12-18', 4, 105),
95
96 (1025, 'Sahil Verma', 'sahil@company.com', 56000, '2022-01-07', 5, 106),
97 (1026, 'Muskan Singh', 'muskan@company.com', 64000, '2020-04-23', 5, 108),
98 (1027, 'Harsh Vardhan', 'harsh@company.com', 61000, '2021-06-30', 5, 106),
99 (1028, 'Nisha Sharma', 'nisha@company.com', 70000, '2019-09-12', 5, 108),
100 (1029, 'Dev Agarwal', 'dev@company.com', 53000, '2023-02-28', 5, 106),
101 (1030, 'Ira Saxena', 'ira@company.com', 78000, '2020-11-16', 5, 108);
102
103
104
105 -- *****
106 -- QUERY 3: AGGREGATION
107 -- Count employees and calculate average salary
108 -- *****
109 SELECT
110     COUNT(*) AS total_employees,
111     AVG(salary) AS average_salary
112 FROM Employee;
113
114

```

STONIN
Input for the program (Optional)

Output:

total_employees	average_salary
30	63833.333333

Feedback New History Save

-- 4. GROUP BY: department-wise employee count and average salary

```

SELECT dept_id,
       COUNT(*) AS employee_count,
       AVG(salary) AS average_salary
FROM Employee
GROUP BY dept_id;

```

byteXL Ctrl + K NISHANT KUMAR 95 ms

queries.sql x MySQL Run

```

80 (1012, 'Ishita Jain', 'ishita@company.com', 57000, '2023-05-15', 2, 103),
81
82 (1013, 'Aditya Patel', 'aditya@company.com', 68000, '2021-02-14', 3, 104),
83 (1014, 'Meera Shah', 'meera@company.com', 75000, '2020-10-19', 3, 104),
84 (1015, 'Rohan Das', 'rohan@company.com', 63000, '2022-03-08', 3, 104),
85 (1016, 'Sneha Iyer', 'sneha@company.com', 71000, '2021-07-13', 3, 104),
86 (1017, 'Manish Yadav', 'manish@company.com', 59000, '2023-04-20', 3, 104),
87 (1018, 'Tanya Kapoor', 'tanya@company.com', 82000, '2019-11-30', 3, 104),
88
89 (1019, 'Nikhil Bansal', 'nikhil@company.com', 54000, '2022-02-21', 4, 105),
90 (1020, 'Kritika Sethi', 'kritika@company.com', 62000, '2021-05-16', 4, 105),
91 (1021, 'Yash Thakur', 'yash@company.com', 57000, '2023-03-11', 4, 105),
92 (1022, 'Aditi Roy', 'aditi@company.com', 66000, '2020-08-09', 4, 105),
93 (1023, 'Mohit Arora', 'mohit@company.com', 59000, '2022-10-14', 4, 105),
94 (1024, 'Diya Khanna', 'diya@company.com', 73000, '2021-12-18', 4, 105),
95
96 (1025, 'Sahil Verma', 'sahil@company.com', 56000, '2022-01-07', 5, 106),
97 (1026, 'Muskan Singh', 'muskan@company.com', 64000, '2020-04-23', 5, 108),
98 (1027, 'Harsh Vardhan', 'harsh@company.com', 61000, '2021-06-30', 5, 106),
99 (1028, 'Nisha Sharma', 'nisha@company.com', 70000, '2019-09-12', 5, 108),
100 (1029, 'Dev Agarwal', 'dev@company.com', 53000, '2023-02-28', 5, 106),
101 (1030, 'Ira Saxena', 'ira@company.com', 78000, '2020-11-16', 5, 108);
102
103
104
105 -- *****
106 -- QUERY 4: GROUP BY
107 -- Department-wise employee count and average salary
108 -- *****
109 SELECT
110     dept_id,
111     COUNT(*) AS employee_count,
112     AVG(salary) AS average_salary
113 FROM Employee
114 GROUP BY dept_id;
115

```

STONIN
Input for the program (Optional)

Output:

dept_id	employee_count	average_salary
1	6	70166.666667
2	6	53833.333333
3	6	69666.666667
4	6	61833.333333
5	6	63666.666667

Feedback New History Save

-- 5. HAVING: departments having more than 5 employees

```

SELECT dept_id,
       COUNT(*) AS employee_count
FROM Employee
GROUP BY dept_id
HAVING COUNT(*) > 5;

```

byteXL Ctrl + K NISHANT KUMAR

queries.sql x MySQL Run 86 ms

```

81
82 (1013, 'Aditya Patel', 'aditya@company.com', 68000, '2021-02-14', 3, 104),
83 (1014, 'Meera Shah', 'meera@company.com', 75000, '2020-10-19', 3, 104),
84 (1015, 'Rohan Das', 'rohan@company.com', 63000, '2022-03-08', 3, 104),
85 (1016, 'Sneha Iyer', 'sneha@company.com', 71000, '2021-07-13', 3, 104),
86 (1017, 'Manish Yadav', 'manish@company.com', 59000, '2023-04-20', 3, 104),
87 (1018, 'Tanya Kapoor', 'tanya@company.com', 82000, '2019-11-30', 3, 104),
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89 (1019, 'Nikhil Bansal', 'nikhil@company.com', 54000, '2022-02-21', 4, 105),
90 (1020, 'Kritika Sethi', 'kritika@company.com', 62000, '2021-05-16', 4, 105),
91 (1021, 'Yash Thakur', 'yash@company.com', 57000, '2023-03-11', 4, 105),
92 (1022, 'Aditi Roy', 'aditi@company.com', 66000, '2020-08-09', 4, 105),
93 (1023, 'Mohit Arora', 'mohit@company.com', 59000, '2022-10-14', 4, 105),
94 (1024, 'Diya Khanna', 'diya@company.com', 73000, '2021-12-18', 4, 105),
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96 (1025, 'Sahil Verma', 'sahil@company.com', 56000, '2022-01-07', 5, 106),
97 (1026, 'Muskan Singh', 'muskan@company.com', 64000, '2020-04-23', 5, 106),
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99 (1028, 'Nisha Sharma', 'nisha@company.com', 70000, '2019-09-12', 5, 106),
100 (1029, 'Dev Agarwal', 'dev@company.com', 53000, '2023-02-28', 5, 106),
101 (1030, 'Ira Saxena', 'ira@company.com', 78000, '2020-11-16', 5, 106);
102
103
104
105 -- QUERY 5: HAVING
106 -- Departments having more than 5 employees
107
108
109 SELECT
110     dept_id,
111     COUNT(*) AS employee_count
112 FROM Employee
113 GROUP BY dept_id
114 HAVING COUNT(*) > 5;
115
116

```

STON Input for the program (Optional)

Output:

dept_id	employee_count
1	6
2	6
3	6
4	6
5	6

Ready Feedback New History Save

-- 6. CASE expression: classify employees according to salary

```

SELECT emp_name,
       salary,
       CASE
           WHEN salary >= 75000 THEN 'High Salary'
           WHEN salary >= 60000 THEN 'Medium Salary'
           ELSE 'Low Salary'
       END AS salary_category
FROM Employee;

```

byteXL Ctrl + K NISHANT KUMAR

queries.sql x MySQL Run 85 ms

```

78 (1010, 'Pooja Nair', 'pooja@company.com', 61000, '2020-12-01', 2, 103),
79 (1011, 'Karan Malhotra', 'karan@company.com', 58000, '2022-07-22', 2, 103),
80 (1012, 'Ishita Jain', 'ishita@company.com', 57000, '2023-05-15', 2, 103),
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82 (1013, 'Aditya Patel', 'aditya@company.com', 68000, '2021-02-14', 3, 104),
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98 (1027, 'Harsh Vardhan', 'harsh@company.com', 61000, '2021-06-30', 5, 106),
99 (1028, 'Nisha Sharma', 'nisha@company.com', 70000, '2019-09-12', 5, 106),
100 (1029, 'Dev Agarwal', 'dev@company.com', 53000, '2023-02-28', 5, 106),
101 (1030, 'Ira Saxena', 'ira@company.com', 78000, '2020-11-16', 5, 106);
102
103
104
105 -- QUERY 6: CASE EXPRESSION
106 -- Classify employees according to salary
107
108
109 SELECT
110     emp_name,
111     salary,
112     CASE
113         WHEN salary >= 75000 THEN 'High Salary'
114         WHEN salary >= 60000 THEN 'Medium Salary'
115         ELSE 'Low Salary'
116     END AS salary_category
117 FROM Employee;
118
119

```

STON Input for the program (Optional)

Output:

emp_name	salary	salary_category
Aarav Sharma	65000.00	Medium Salary
Riya Verma	72000.00	Medium Salary
Arjun Singh	58000.00	Low Salary
Neha Gupta	81000.00	High Salary
Kabir Mehta	69000.00	Medium Salary
Ananya Rao	76000.00	High Salary
Rahul Kumar	48000.00	Low Salary
Simran Kaur	52000.00	Low Salary
Vikram Joshi	55000.00	Low Salary
Pooja Nair	61000.00	Medium Salary
Karan Malhotra	58000.00	Low Salary
Ishita Jain	57000.00	Low Salary
Aditya Patel	68000.00	Medium Salary
Meera Shah	75000.00	High Salary
Rohan Das	63000.00	Medium Salary
Sneha Iyer	71000.00	Medium Salary
Manish Yadav	59000.00	Low Salary
Tanya Kapoor	82000.00	High Salary
Nikhil Bansal	54000.00	Low Salary
Kritika Sethi	62000.00	Medium Salary
Yash Thakur	57000.00	Low Salary
Aditi Roy	66000.00	Medium Salary
Mohit Arora	59000.00	Low Salary
Diya Khanna	73000.00	Medium Salary
Sahil Verma	56000.00	Low Salary
Muskan Singh	64000.00	Medium Salary
Harsh Vardhan	61000.00	Medium Salary
Nisha Sharma	70000.00	Medium Salary
Dev Agarwal	53000.00	Low Salary
Ira Saxena	78000.00	High Salary

Ready Feedback New History Save

-- 7. ORDER BY: display employees in descending order of salary

```

SELECT emp_name, salary
FROM Employee
ORDER BY salary DESC;

```

byteXL Ctrl + K NISHANT KUMAR

queries.sql x + MySQL Run

```

74
75 (1007, 'Rahul Kumar', 'rahul@company.com', 48000, '2022-04-11', 2, 103),
76 (1008, 'Simeran Kaur', 'simeran@company.com', 52000, '2023-01-25', 2, 103),
77 (1009, 'Vikram Joshi', 'vikram@company.com', 55000, '2021-09-17', 2, 103),
78 (1010, 'Pooja Nair', 'pooja@company.com', 61000, '2022-12-01', 2, 103),
79 (1011, 'Karan Malhotra', 'karan@company.com', 58000, '2022-07-22', 2, 103),
80 (1012, 'Ishita Jain', 'ishita@company.com', 57000, '2023-05-15', 2, 103),
81
82 (1013, 'Aditya Patel', 'aditya@company.com', 68000, '2021-02-14', 3, 104),
83 (1014, 'Meera Shah', 'meera@company.com', 75000, '2020-10-19', 3, 104),
84 (1015, 'Rohan Das', 'rohan@company.com', 63000, '2022-03-08', 3, 104),
85 (1016, 'Sneha Iyer', 'sneha@company.com', 71000, '2021-07-13', 3, 104),
86 (1017, 'Manish Yadav', 'manish@company.com', 59000, '2023-04-20', 3, 104),
87 (1018, 'Tanya Kapoor', 'tanya@company.com', 82000, '2019-11-30', 3, 104),
88
89 (1019, 'Nikhil Bansal', 'nikhil@company.com', 54000, '2022-02-21', 4, 105),
90 (1020, 'Kritika Sethi', 'kritika@company.com', 62000, '2021-05-16', 4, 105),
91 (1021, 'Yash Thakur', 'yash@company.com', 57000, '2023-03-11', 4, 105),
92 (1022, 'Aditi Roy', 'aditi@company.com', 66000, '2020-08-09', 4, 105),
93 (1023, 'Mohit Arora', 'mohit@company.com', 59000, '2022-10-14', 4, 105),
94 (1024, 'Diya Khanna', 'diya@company.com', 73000, '2021-12-18', 4, 105),
95
96 (1025, 'Sahil Verma', 'sahil@company.com', 56000, '2022-01-07', 5, 106),
97 (1026, 'Muskan Singh', 'muskan@company.com', 64000, '2020-04-23', 5, 106),
98 (1027, 'Harsh Vardhan', 'harsh@company.com', 61000, '2021-06-30', 5, 106),
99 (1028, 'Nisha Sharma', 'nisha@company.com', 70000, '2019-09-12', 5, 106),
100 (1029, 'Dev Agarwal', 'dev@company.com', 53000, '2023-02-28', 5, 106),
101 (1030, 'Ira Saxena', 'ira@company.com', 78000, '2020-11-16', 5, 106),
102
103
104 --
105 -- QUERY 7: ORDER BY
106 -- Display employees in descending order of salary
107 --
108
109 SELECT
110     emp_name,
111     salary
112 FROM Employee
113 ORDER BY salary DESC;
114
115

```

STON

Input for the program (Optional)

Output:

emp_name	salary
Tanya Kapoor	82000.00
Mehta Gupta	81800.00
Ira Saxena	78000.00
Ananya Rao	76000.00
Meera Shah	75000.00
Diya Khanna	73000.00
Riya Verma	72000.00
Sneha Iyer	71000.00
Nisha Sharma	70000.00
Kabir Mehta	69000.00
Aditya Patel	68000.00
Aditi Roy	66000.00
Aarav Sharma	65000.00
Muskan Singh	64000.00
Rohan Das	63000.00
Kritika Sethi	62000.00
Pooja Nair	61000.00
Harsh Vardhan	61000.00
Manish Yadav	59000.00
Mohit Arora	59000.00
Arjun Singh	58000.00
Yash Thakur	57000.00
Ishita Jain	57000.00
Sahil Verma	56000.00
Vikram Joshi	55000.00
Nikhil Bansal	54000.00
Dev Agarwal	53000.00
Simeran Kaur	52000.00
Karan Malhotra	50000.00
Rahul Kumar	48000.00

Light 13

Feedback New History Save

```

-- 8. Department-wise total salary
SELECT dept_id,
       SUM(salary) AS total_salary
FROM Employee
GROUP BY dept_id
ORDER BY total_salary DESC;

```

byteXL Ctrl + K NISHANT KUMAR

queries.sql x + MySQL Run

```

80 (1012, 'Ishita Jain', 'ishita@company.com', 57000, '2023-05-15', 2, 103),
81
82 (1013, 'Aditya Patel', 'aditya@company.com', 68000, '2021-02-14', 3, 104),
83 (1014, 'Meera Shah', 'meera@company.com', 75000, '2020-10-19', 3, 104),
84 (1015, 'Rohan Das', 'rohan@company.com', 63000, '2022-03-08', 3, 104),
85 (1016, 'Sneha Iyer', 'sneha@company.com', 71000, '2021-07-13', 3, 104),
86 (1017, 'Manish Yadav', 'manish@company.com', 59000, '2023-04-20', 3, 104),
87 (1018, 'Tanya Kapoor', 'tanya@company.com', 82000, '2019-11-30', 3, 104),
88
89 (1019, 'Nikhil Bansal', 'nikhil@company.com', 54000, '2022-02-21', 4, 105),
90 (1020, 'Kritika Sethi', 'kritika@company.com', 62000, '2021-05-16', 4, 105),
91 (1021, 'Yash Thakur', 'yash@company.com', 57000, '2023-03-11', 4, 105),
92 (1022, 'Aditi Roy', 'aditi@company.com', 66000, '2020-08-09', 4, 105),
93 (1023, 'Mohit Arora', 'mohit@company.com', 59000, '2022-10-14', 4, 105),
94 (1024, 'Diya Khanna', 'diya@company.com', 73000, '2021-12-18', 4, 105),
95
96 (1025, 'Sahil Verma', 'sahil@company.com', 56000, '2022-01-07', 5, 106),
97 (1026, 'Muskan Singh', 'muskan@company.com', 64000, '2020-04-23', 5, 106),
98 (1027, 'Harsh Vardhan', 'harsh@company.com', 61000, '2021-06-30', 5, 106),
99 (1028, 'Nisha Sharma', 'nisha@company.com', 70000, '2019-09-12', 5, 106),
100 (1029, 'Dev Agarwal', 'dev@company.com', 53000, '2023-02-28', 5, 106),
101 (1030, 'Ira Saxena', 'ira@company.com', 78000, '2020-11-16', 5, 106),
102
103
104 --
105 -- QUERY 8: DEPARTMENT-WISE TOTAL SALARY
106 --
107
108 SELECT
109     dept_id,
110     SUM(salary) AS total_salary
111 FROM Employee
112 GROUP BY dept_id
113 ORDER BY total_salary DESC;
114
115

```

STON

Input for the program (Optional)

Output:

dept_id	total_salary
1	421000.00
3	418000.00
5	382000.00
4	371000.00
2	323000.00

Light 13

Feedback New History Save

```

-- 9. Project-wise employee count
SELECT project_id,
       COUNT(*) AS employee_count
FROM Employee
GROUP BY project_id
ORDER BY employee_count DESC;

```

byteXL Ctrl + K NISHANT KUMAR 103 ms

queries.sql x + MySQL Run

```

81
82 (1013, 'Aditya Patel', 'aditya@company.com', 68000, '2021-02-14', 3, 104),
83 (1014, 'Meera Shah', 'meera@company.com', 75000, '2020-10-19', 3, 104),
84 (1015, 'Rohan Das', 'rohan@company.com', 63000, '2022-03-08', 3, 104),
85 (1016, 'Sneha Iyer', 'sneha@company.com', 71000, '2021-07-13', 3, 104),
86 (1017, 'Manish Yadav', 'manish@company.com', 59000, '2023-04-20', 3, 104),
87 (1018, 'Tanya Kapoor', 'tanya@company.com', 82000, '2019-11-30', 3, 104),
88
89 (1019, 'Nikhil Bansal', 'nikhil@company.com', 54000, '2022-02-21', 4, 105),
90 (1020, 'Kritika Sethi', 'kritika@company.com', 62000, '2021-05-16', 4, 105),
91 (1021, 'Yash Thakur', 'yash@company.com', 57000, '2023-03-11', 4, 105),
92 (1022, 'Aditi Roy', 'aditi@company.com', 66000, '2020-08-09', 4, 105),
93 (1023, 'Mohit Arora', 'mohit@company.com', 59000, '2022-10-14', 4, 105),
94 (1024, 'Diya Khanna', 'diya@company.com', 73000, '2021-12-18', 4, 105),
95
96 (1025, 'Sahil Verma', 'sahil@company.com', 56000, '2022-01-07', 5, 106),
97 (1026, 'Muskan Singh', 'muskan@company.com', 64000, '2020-04-23', 5, 106),
98 (1027, 'Harsh Vardhan', 'harsh@company.com', 61000, '2021-06-30', 5, 106),
99 (1028, 'Nisha Sharma', 'nisha@company.com', 70000, '2019-09-12', 5, 106),
100 (1029, 'Dev Agarwal', 'dev@company.com', 53000, '2023-02-28', 5, 106),
101 (1030, 'Ira Saxena', 'ira@company.com', 78000, '2020-11-16', 5, 106);
102
103
104
105 -- *****
106 -- QUERY 9: PROJECT-WISE EMPLOYEE COUNT
107 -- *****
108
109 SELECT
110     project_id,
111     COUNT(*) AS employee_count
112 FROM Employee
113 GROUP BY project_id
114 ORDER BY employee_count DESC;
115
116

```

Ready Feedback New History Save Light

STDIN

Input for the program (Optional)

Output:

project_id	employee_count
103	6
104	6
105	6
106	3
108	3
101	2
102	2
107	2

-- 10. Display employees with department and project details

```

SELECT e.emp_name,
       d.dept_name,
       p.project_name,
       e.salary
FROM Employee e
JOIN Department d ON e.dept_id = d.dept_id
JOIN Project p ON e.project_id = p.project_id
ORDER BY e.salary DESC;

```

byteXL Ctrl + K NISHANT KUMAR 69 ms

queries.sql x + MySQL Run

```

81
82 (1013, 'Aditya Patel', 'aditya@company.com', 68000, '2021-02-14', 3, 104),
83 (1014, 'Meera Shah', 'meera@company.com', 75000, '2020-10-19', 3, 104),
84 (1015, 'Rohan Das', 'rohan@company.com', 63000, '2022-03-08', 3, 104),
85 (1016, 'Sneha Iyer', 'sneha@company.com', 71000, '2021-07-13', 3, 104),
86 (1017, 'Manish Yadav', 'manish@company.com', 59000, '2023-04-20', 3, 104),
87 (1018, 'Tanya Kapoor', 'tanya@company.com', 82000, '2019-11-30', 3, 104),
88
89 (1019, 'Nikhil Bansal', 'nikhil@company.com', 54000, '2022-02-21', 4, 105),
90 (1020, 'Kritika Sethi', 'kritika@company.com', 62000, '2021-05-16', 4, 105),
91 (1021, 'Yash Thakur', 'yash@company.com', 57000, '2023-03-11', 4, 105),
92 (1022, 'Aditi Roy', 'aditi@company.com', 66000, '2020-08-09', 4, 105),
93 (1023, 'Mohit Arora', 'mohit@company.com', 59000, '2022-10-14', 4, 105),
94 (1024, 'Diya Khanna', 'diya@company.com', 73000, '2021-12-18', 4, 105),
95
96 (1025, 'Sahil Verma', 'sahil@company.com', 56000, '2022-01-07', 5, 106),
97 (1026, 'Muskan Singh', 'muskan@company.com', 64000, '2020-04-23', 5, 106),
98 (1027, 'Harsh Vardhan', 'harsh@company.com', 61000, '2021-06-30', 5, 106),
99 (1028, 'Nisha Sharma', 'nisha@company.com', 70000, '2019-09-12', 5, 106),
100 (1029, 'Dev Agarwal', 'dev@company.com', 53000, '2023-02-28', 5, 106),
101 (1030, 'Ira Saxena', 'ira@company.com', 78000, '2020-11-16', 5, 106);
102
103
104
105 -- *****
106 -- QUERY 10: EMPLOYEE + DEPARTMENT + PROJECT DETAILS
107 -- *****
108
109 SELECT
110     e.emp_name,
111     d.dept_name,
112     p.project_name,
113     e.salary
114 FROM Employee e
115 JOIN Department d
116     ON e.dept_id = d.dept_id
117 JOIN Project p
118     ON e.project_id = p.project_id
119 ORDER BY e.salary DESC;
120
121
122

```

Ready Feedback New History Save Light

STDIN

Input for the program (Optional)

Output:

emp_name	dept_name	project_name	salary
Tanya Kapoor	Finance	Financial Analytics	82000.00
Neha Gupta	Computer Science	Cloud Migration	81000.00
Ira Saxena	Operations	Business Automation	78000.00
Ananya Rao	Computer Science	Cloud Migration	76000.00
Meera Shah	Finance	Financial Analytics	75000.00
Diya Khanna	Marketing	Digital Marketing	73000.00
Riya Verma	Computer Science	Web Development	72000.00
Sneha Iyer	Finance	Financial Analytics	71000.00
Nisha Sharma	Operations	Business Automation	70000.00
Kabir Mehta	Computer Science	Web Development	69000.00
Aditya Patel	Finance	Financial Analytics	68000.00
Aditi Roy	Marketing	Digital Marketing	66000.00
Aarav Sharma	Computer Science	AI Research	65000.00
Muskan Singh	Operations	Business Automation	64000.00
Rohan Das	Finance	Financial Analytics	63000.00
Kritika Sethi	Marketing	Digital Marketing	62000.00
Pooja Nair	Human Resources	Recruitment Portal	61000.00
Harsh Vardhan	Operations	Inventory System	61000.00
Manish Yadav	Finance	Financial Analytics	59000.00
Mohit Arora	Marketing	Digital Marketing	59000.00
Arjun Singh	Computer Science	AI Research	58000.00
Yash Thakur	Marketing	Digital Marketing	57000.00
Ishita Jain	Human Resources	Recruitment Portal	57000.00
Sahil Verma	Operations	Inventory System	56000.00
Vikram Joshi	Human Resources	Recruitment Portal	55000.00
Nikhil Bansal	Marketing	Digital Marketing	54000.00
Dev Agarwal	Operations	Inventory System	53000.00
Sirvan Kaur	Human Resources	Recruitment Portal	52000.00
Karan Malhotra	Human Resources	Recruitment Portal	50000.00
Rahul Kumar	Human Resources	Recruitment Portal	48000.00